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https://www.tensorflow.org/api_docs/python/tf/compat/v1/VariableAggregation

> *Indicates how a distributed variable will be aggregated.*

Function Signatures

```
var.assign_add(x)
```

Code Examples

```
strategy = tf.distribute.MirroredStrategy(["GPU:0", "GPU:1"])
with strategy.scope():
v = tf.Variable(5.0, aggregation=tf.VariableAggregation.MEAN)
@tf.function
def update_fn():
return v.assign_add(1.0)
strategy.run(update_fn)
PerReplica:{
0: ,
1:
}
```

```
strategy = tf.distribute.MirroredStrategy(["GPU:0", "GPU:1"])
```

```
with strategy.scope():
```

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v = tf.Variable(5.0, aggregation=tf.VariableAggregation.MEAN)
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@tf.function
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def update_fn():
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```
return v.assign_add(1.0)
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```
strategy.run(update_fn)
```

Documentation

Indicates how a distributed variable will be aggregated.

tf.distribute.Strategy distributes a model by making multiple copies (called "replicas") acting on different elements of the input batch in a data parallel model. When performing some variable-update operation, for example var.assign_add(x), in a model, we need to resolve how to combine the different values for x computed in the different replicas.

- NONE: This is the default, giving an error if you use a

variable-update operation with multiple replicas.

- SUM: Add the updates across replicas.
- MEAN: Take the arithmetic mean ("average") of the updates across replicas.
- ONLY_FIRST_REPLICA: This is for when every replica is performing the same

update, but we only want to perform the update once. Used, e.g., for the global step counter.

For example:

• ONLY_FIRST_TOWER: Deprecated alias for ONLY_FIRST_REPLICA.

Class Variables